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DEPARTMENT: CYBER SECURITY

FACULTY: COMPUTING

COURSE CODE: COS 202

COURSE TITLE: COMPUTER PROGRAMING II(PYTHON)

<u>Assignment</u>

1. Using PYTHON programming language, develop an interactive and well-designed screen MATHEMATICAL CALCULATOR (MC) program containing components such as +, -, /, *, ^, %, C (Clear), OFF etc. (NB: It’s a simple calculator not a complex scientific one.)

Answer

```
print(“=========================================”)

print(“SIMPLE MATHEMATICAL CALCULATOR”)

print(“=========================================”)


while True:

   print(“1. Addition (+)”)

   print(“2. Subtraction (-)”)

   print(“3. Multiplication (*)”)

   print(“4. Division (/)”)

   print(“5. Modulus (%)”)

   print(“6. Power (^)”)

   print(“7. Clear ©”)

   print(“8. OFF”)

   choice = input(“Enter your choice (1-8): “)

   match choice:

      case “1”:

         print(“You selected Addition”)

         num1 = int(input(“Enter first number: “))
```

```
    num2 = int(input(“Enter second number: “))
    addition = num1 + num2
    print(“Addition =”, addition)
case “2”:
    Print(“You selected Subtraction”)
    num1 = int(input(“Enter first number: “))
    num2 = int(input(“Enter second number: “))
    subtraction = num1 – num2
    print(“Subtraction =”, subtraction)
case “3”:
    print(“You selected Multiplication”)
    num1 = int(input(“Enter first number: “))
    num2 = int(input(“Enter second number: “))
    multiplication = num1 * num2
    print(“Multiplication =”, multiplication)
case “4”:
    print(“You selected Division”)
    num1 = int(input(“Enter first number: “))
    num2 = int(input(“Enter second number: “))
    if num2 == 0:
        print(“Division by zero is not allowed.”)
    else:
        division = num1 / num2
        print(“Division =”, division)
```

```
    case “5”:
        print(“You selected Modulus”)
        num1 = int(input(“Enter first number: “))
        num2 = int(input(“Enter second number: “))
        modulus = num1 % num2
        print(“Modulus =”, modulus)
    case “6”:
        print(“You selected Power”)
        num1 = int(input(“Enter first number: “))
        num2 = int(input(“Enter second number: “))
        power = num1 ** num2
        print(“Power =”, power)
    case “7”:
        print(“Calculator Cleared!”)
    case “8”:
        print(“Calculator OFF”)
        print(“Thank you for using the calculator.”)
        break
    case _:
        print(“Invalid choice. Please try again.”)
```

Output

SIMPLE MATHEMATICAL CALCULATOR

1. Addition (+)
2. Subtraction (-)

3. Multiplication (*)

4. Division (/)

5. Modulus (%)

6. Power (^)

7. Clear ©

8. OFF

Enter your choice (1-8): 4

You selected Division

Enter first number: 45

Enter second number: 5

Division = 9.0

1. Addition (+)

2. Subtraction (-)

3. Multiplication (*)

4. Division (/)

5. Modulus (%)

6. Power (^)

7. Clear ©

8. OFF

Enter your choice (1-8): 5

You selected Modulus

Enter first number: 13

Enter second number: 4

Modulus = 1

1. Addition (+)

2. Subtraction (-)

3. Multiplication (*)

4. Division (/)

5. Modulus (%)

6. Power (^)

7. Clear ©

8. OFF

Enter your choice (1-8): 8

Calculator OFF

Thank you for using the calculator.

[Program finished]

2. CGPA is not something that should constrain you to visiting AVERS at all time. Using the powerful provision of selection control statements in PYTHON, develop a PERSONAL POCKET CGPA Calculator (PPC).

Answer

print("==")

print("PERSONAL POCKET CGPA CALCULATOR")

print("==")

total_grade_point = float(input("Enter Total Grade Point: "))

total_credit_unit = float(input("Enter Total Credit Unit: "))

```
    cgpa = total_grade_point / total_credit_unit
    print(“CGPA =”, cgpa)
    if cgpa >= 4.50 and cgpa <= 5.00:
        print(“Class of Degree: First Class”)
    elif cgpa >= 3.50 and cgpa <= 4.49:
        print(“Class of Degree: Second Class Upper”)
    elif cgpa >= 2.40 and cgpa <= 3.49:
        print(“Class of Degree: Second Class Lower”)
    elif cgpa >= 1.50 and cgpa <= 2.39:
        print(“Class of Degree: Third Class”)
    elif cgpa >= 1.00 and cgpa <= 1.49:
        print(“Class of Degree: Pass”)
    else:
        print(“Class of Degree: Fail”)
```

Output

```
================================================
PERSONAL POCKET CGPA CALCULATOR
================================================
Enter Total Grade Point: 69.56
Enter Total Credit Unit: 21
CGPA = 3.3123809523809524
Class of Degree: Second Class Lower


[Program finished]
```

The programs were successfully developed using Python, tested, and uploaded to GitHub.